Handling Data 4 — Experimental and theoretical probability · Sample spaces

Give probabilities as fractions in their lowest terms unless asked otherwise. Marks in brackets. ★ = harder question.

1 Each face of a cube is painted red or blue: two faces red (R), four blue (B). The cube is rolled 12 times: B B R B B B R B B R B B. [7]

a Plot the relative frequency of red after 1, 2, 3, … 12 rolls on a graph.

b From the 12 rolls, estimate the probability of **(i)** a red face, **(ii)** a blue face.

c Comment on your results.

2 A card is picked at random from an ordinary pack of 52. Find the probability that it is [5]

a red **b** a spade **c** not a queen **d** a black ace.

3 A letter is chosen at random from the word ANTIDISESTABLISHMENTARIANISM (28 letters). Find the probability that it is [4]

a A **b** I **c** Z **d** a vowel.

4 Two fair dice are thrown and the score is the *difference* between the two numbers (the larger minus the smaller). [9]

a Copy and complete the sample space showing all 36 outcomes.

–	1	2	3	4	5	6
1	0	1				
2		0		2		
3						
4						
5						
6						

b Find the probability that the score is **(i)** 0, **(ii)** at least 4, **(iii)** odd, **(iv)** a prime number.

c The game is played 72 times. How many odd scores would you expect?

Total: 25 marks

Check: theoretical probability needs *equally likely* outcomes · $P(\text{not } E) = 1 - P(E)$ · a sample space lists every outcome once — count carefully · expected frequency = probability × number of trials.